

PCWD ED PATHWAYS - UPDATED 07.23.2024

Andrea W Thorp



Pediatric Emergency Department Pathways

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Table of Contents

<u>Asthma</u>	2
<u>Bronchiolitis</u>	4
<u>Community – Acquired Pneumonia</u>	6
<u>Dehydration Due to Gastroenteritis</u>	9
<u>Status Epilepticus</u>	10
<u>Seizure</u>	11
<u>Croup</u>	12

Emergency Department Asthma Pathway

Child with Asthma

Criteria:

- Wheezing
- Child > 12 months
- History of asthma or previous wheezing
- History of responding or currently responds to Albuterol treatments

*Asthma has variability among children: must complete an integrated overall assessment to decide severity

Assess Severity

Mild

- Mild wheeze
- Minimal retractions
- Mild prolonged expiratory phase
- Good air movement
- Saturations > 94%
- Short of breath with agitation/activity
- Talking full sentences

Moderate

- Expiratory wheeze
- Intercostal retractions
- Prolonged expiratory phase
- Decreased air movement
- Short of breath with rest
- Talking in phrases

Severe

- Inspiratory / Expiratory wheeze
- Retractions/Abdominal breathing
- Prolonged expiratory phase
- Fair/Poor air movement
- Talking single words

Imminent Respiratory Failure

- Absent wheeze
- Shallow breaths/Tired
- Poor/Absent air movement

Evaluation

Chest Xray, not routinely recommended unless:

- Severe symptoms
- Evaluate for other diagnoses
- Asymmetric exam

Respiratory viral panel not routinely recommended

See Peds Asthma ED Order set for dosing

- Oral steroid: *
Dexamethasone or prednisolone
- Albuterol MDI (with spacer/mask) or single Albuterol med neb

- Oral Dexamethasone* (IV formulation given orally)
- Albuterol nebulized every 30 mins x 3
- +/-Atrovent nebulized (Can be given X1 or X3)
- Oxygen supplement if saturations < 90% while awake (88% while asleep)

*Avoid re-dosing if given in clinic

- Dexamethasone (Prefer IV/IM)*
- Albuterol given either:
 - o Nebulized every 30 mins x3
 - o Continuous aerosolized breathing treatment dose = 0.5mg/kg/hour with a max between 20-40mg
- Atrovent nebulized with Albuterol
- Oxygen supplement if saturations <90% (88% while asleep)
- Magnesium IV if severe or refractory
- Epinephrine IM if critically ill
- Start oxygen support with High flow NC or NIPPV. Emergent intubation as a last resort
- NS bolus if dehydrated

*Avoid re-dosing if given in clinic

Re-assess after 1 hour

- Repeat treatments if necessary.



Re-assess after 2nd hour

- Decide disposition



Meets Discharge Criteria:

- Normal oxygen saturations and respiratory rate for age
- Oral steroids for 4 more days
- Review new/home asthma meds
- Case management: Med neb machine delivered to bedside if indicated

Admission:

Contact Pediatric Hospitalist for site-specific criteria to decide Pediatric floor versus PICU level of care

Continue nebulized treatments on a scheduled basis until transfer arranged

Bronchiolitis Emergency Department Pathway

High Risk Criteria:

- <12 weeks, history of prematurity, underlying cardiac and pulmonary disease, immunodeficiency
- Infants evaluated early in the course of illness typically will worsen if gestational age <34 weeks, RR >70 and <3 months of age

Criteria:

- Typically < 2 y/o
 - Clinical picture consistent with bronchiolitis (viral URI with increased work of breathing, wheezing, coarse breath sounds, increased respiratory rate)
- > 2 years old will overlap with asthma pathway

Exclude:

- Cardiac disease
- History of airway defects or chronic lung disease
- Immunodeficiency

Assess Severity

Mild

- Assess hydration
- Mild tachypnea for age
- Mild work of breathing
- Normal mental status
- Mild wheeze/clear lungs on room air



Nasal suction (with NS drops and mushroom tip suctioning catheter attached to wall suction)



Moderate

- Assess hydration
- Moderate tachypnea for age
- Retractions
- Fussy, anxious
- Crackles, wheeze
- < 1.5 liters oxygen supplementation



- Nasal suction (with NS drops and mushroom tip suction catheter attached to wall suction)
- Bronchodilators not routinely recommended but can be used as a trial (document response)
- Alternative treatments: Racemic epinephrine, Hypertonic normal saline med neb, High-flow NC **see attached below*

Re-assess response.

Severe

- Assess hydration
- Severe tachypnea for age
- Nasal flaring, grunting, head bobbing
- Inconsolable, listless
- Poor air movement, wheeze, crackles
- > 1.5 liters oxygen



Meets discharge criteria*

- Oxygen saturations $\geq 90\%$ while awake & $\geq 88\%$ while asleep and feeding
 - Reliable caregiver
 - Follow-Up can be arranged
 - Supportive care: suctioning, feeding well, fever management
 - ED case management: med neb machine if needed delivered to bedside
- *see high-risk infant criteria above

Admission:

- Does not meet discharge criteria
- Consult Pediatric Hospitalist for site-specific criteria to decide pediatric floor versus PICU level of care

Additional Treatment Considerations

May be considered in infants with significant respiratory distress that is not alleviated sufficiently by supportive care interventions.

Albuterol	Studies have NOT demonstrated a consistent benefit for albuterol treatment in infants with typical bronchiolitis . An albuterol trial may be considered in children with features suggestive of possible asthma (recurrent wheezing, age > 12 months, prior inhaled corticosteroid use).			
	Albuterol should NOT be continued if the patient does not respond to test dose. If responsive to albuterol test dose, may continue: <table> <tr> <td>MDI</td><td>Frequency of every 1-4 hours as needed</td></tr> <tr> <td>Intermittent nebulization</td><td>Frequency of every 2-6 hours as needed</td></tr> </table>	MDI	Frequency of every 1-4 hours as needed	Intermittent nebulization
MDI	Frequency of every 1-4 hours as needed			
Intermittent nebulization	Frequency of every 2-6 hours as needed			
Racemic Epinephrine	α and β adrenergic agonist Consider use in patients with increasing severe respiratory distress			
HFNC	Provides warmed, humidified air with adjustable oxygen concentration Reduces WOB Consider if $FiO_2 > 40\%$			
Antibiotics	Do NOT prescribe antibiotics without evidence of bacterial infection: Bacterial superinfections are uncommon in this age group. Prescribe antibiotics only when there is a specific secondary infection			
Hypertonic Saline	Regular administration of nebulized hypertonic (3%) saline to inpatients has been shown to shorten length of stay by one day. Benefit comes with regular use every 6-8 hours for the entire hospitalization. Consider initiating a hypertonic saline regimen (standing every 6-8 hours) in children with a prolonged hospital stay or those not showing improvement over the first 48 hours of admission.			

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The clinical pathways are based upon publicly available medical evidence and/or a consensus of medical practitioners at The Children's Hospital of Philadelphia ("CHOP") and are current at the time of publication. These clinical pathways are intended to be a guide for practitioners and may need to be adapted for each specific patient based on the practitioner's professional judgment, consideration of any unique circumstances, the needs of each patient and their family, and/or the availability of various resources at the health care institution where the patient is located.

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Emergency Department Pathway Community – Acquired Pneumonia

Child with suspected or confirmed pneumonia

Criteria:

Suspected or confirmed pneumonia

Exclude:

- Cystic fibrosis or chronic lung disease
- Immunodeficiency
- Tracheostomy
- Hospitalized pneumonia
- Aspiration pneumonia



Assess Severity

Mild

- >3 months
- Saturations >90%
- Tolerating oral intake
- Adequate follow-up

Moderate

- <3 months
- Moderate respiratory distress
- Saturations < 90%
- Not tolerating oral intake
- No follow-up
- Failed oral antibiotics after 48 hours

Severe

* Includes all/any moderate criteria

- Severe respiratory distress
- Mechanical support of airway
- Altered mental status



Evaluation

- CXR: Not needed unless diagnosis unsure
- Labs: Not needed



Discharge Criteria:

- Respiratory effort normal
- Fever resolved/trending to normal
- No oxygen requirement
- Tolerating oral intake
- Prescribe oral antibiotics (see separate sheet)



Evaluation

- CXR+CBC recommended
- RVP
- MRSA nasal surveillance swab
- Blood culture if ill appearing



Admission:

- Does not meet discharge criteria
- IV antibiotics (see pediatric pneumonia order set)
- Support airway as appropriate



Effusion on CXR:

- Mild: obtain lateral decubitus XR
- Moderate/Severe: Consult Pediatrics

Refer to "Complicated Pneumonia Pathway"

Initial Antibiotic Recommendations for Patients with Community-Acquired Pneumonia

	First-Line Therapy	β -lactam Allergy ¹	Duration of Tx/Comments
Mild Pneumonia* (Outpatient) *see atypical pneumonia below	Amoxicillin 90 mg/kg/day, divided Q8 - Q12 hrs; max dose: 4 grams/day	1st Line Recommendation: Clindamycin (PO) 30 mg/kg/day divided Q8 hrs; max: 1.8 g/day OR 2nd Line Recommendation: Levofloxacin (PO) , 6mo and < 5 yrs: 20 mg/kg/day, divided Q12 hrs , 5yr: 10 mg/kg/day, Q24 hrs; max: 500 mg	Duration: 7 days Target pathogen: <i>Streptococcus pneumoniae</i> High dose amoxicillin active against most <i>S. pneumoniae</i> Clindamycin active against ~ 90% of <i>S. pneumoniae</i> ; Levofloxacin active against > 95% of <i>S. pneumoniae</i> , <i>Mycoplasma pneumoniae</i> , and <i>Legionella</i> ; Oral cephalosporins inferior to high-dose amoxicillin for <i>S. pneumoniae</i> ; Azithromycin resistance in up to 40% of <i>S. pneumoniae</i> and therefore these agents are not recommended.
Moderate Pneumonia (Inpatient)	Ampicillin 300 mg/kg/day divided Q6hrs; max: 2g/dose OR Consider Ceftriaxone monotherapy for unimmunized patients 100 mg/kg/day, Q24 hrs; max 2g/dose	1st Line Recommendation: Clindamycin (IV) 40 mg/kg/day, divided Q8 hrs; max: 2.7 g/day OR 2nd Line Recommendation: Levofloxacin (IV) , 6mo and < 5 yrs: 20 mg/kg/day, divided Q12 hrs , 5yr: 10mg/kg/day, Q24 hrs; max: 500 mg	Duration: 7 days, or at least 48 hours from resolution of fever and tachypnea (whichever is longer), including oral/outpatient therapy. Target pathogen: <i>S. pneumoniae</i> High dose Ampicillin active against most <i>S. pneumoniae</i> Clindamycin active against ~ 90% of <i>S. pneumoniae</i> Levofloxacin active against > 95% of <i>S. pneumoniae</i> , <i>Mycoplasma pneumoniae</i> , and <i>Legionella</i> Azithromycin resistance in up to 40% of <i>S. pneumoniae</i> and therefore not recommended.
Complicated: Pneumonia (Inpatient)	Clindamycin (IV) 40 mg/kg/day, div Q8hrs; max: 2.7 g/day *AND* Ceftriaxone 100 mg/kg/day, Q24 hrs; max: 2 g/dose	Clindamycin (IV) 40 mg/kg/day, divided Q8 hrs; max: 2.7 g/day *AND* Levofloxacin (IV) , 6mo and < 5 yrs: 20 mg/kg/day, divided Q12 hrs , 5yr: 10 mg/kg/day, Q24 hrs; max: 500 mg	Duration: 7 days from drainage of effusion. If not amenable to drainage, 7 days from afebrile. Target pathogens: <i>Streptococcus pneumoniae</i> , <i>Streptococcus pyogenes</i> (Group A Strep), <i>S. aureus</i> Clindamycin for suspected CA-MRSA, active against ~ 85% of MRSA (CA and HA)
Severe/Complicated Pneumonia (ICU)²	Vancomycin (IV) 15 mg/kg/dose (max 500 mg), Q6 hrs *AND* Ceftriaxone 100 mg/kg/day, Q24 hrs; max: 2 g/dose	Vancomycin (IV) 15 mg/kg/dose (max 500 mg), Q6 hrs. *AND* Levofloxacin (IV) , 6mo and < 5 yrs: 20 mg/kg/day, divided Q12 hrs , 5yr: 10 mg/kg/day, Q24 hrs; max: 500 mg	Duration: 7 days from afebrile. Target pathogens: <i>S. pneumoniae</i> , <i>S. pyogenes</i> , <i>S. aureus</i> <i>Vancomycin for suspected CA-MRSA in severe or life-threatening conditions</i>
Atypical Pneumonia (all locations and severities)³ * *Children >5 years	1st Line (in addition to treatment(s) recommended above) – see Comments: Azithromycin 10 mg/kg on day 1, single dose (max: 500 mg/day), followed by 5 mg/kg/day, once daily Q24 hrs, thereafter (max: 250 mg/day) OR Levofloxacin (IV/PO) , 6mo and < 5 yrs: 20 mg/kg/day, div Q12 hrs , 5yr: 10 mg/kg/day, once daily Q24 hrs; max: 500 mg		Target pathogens: <i>Mycoplasma pneumoniae</i> <i>Legionella pneumophila</i> Duration (for proven infections): <i>Mycoplasma</i> Azithromycin: 5 days Levofloxacin: 10 days <i>Legionella pneumophila:</i> Azithromycin: 5-10 days Levofloxacin: 14-21 days Azithromycin does not treat <i>S. pneumoniae</i> or <i>S. aureus</i>; please administer in addition to drugs above for typical CAP. Levofloxacin has activity against <i>S. pneumoniae</i> as well as atypicals so no additional agents are needed when levofloxacin otherwise indicated.

If increasing respiratory distress, increasing respiratory support requirement, or worsening fever curve after > 48 hours of preferred first line therapy, consider [Treatment Failure](#).

Footnotes

1. Defined by urticaria or anaphylaxis (Type I mediated hypersensitivity).
2. Includes pneumonia with moderate-large parapneumonic effusion.
3. Includes patients hospitalized in the ICU who are critically ill from suspected bacterial pneumonia.
4. Atypical pneumonia (often characterized by non-lobar, patchy, or interstitial pattern on CXR; insidious onset; low-grade of fever, malaise, H/A, cough; minimal auscultatory findings relative to CXR) is often caused by respiratory viruses, but may be caused by atypical bacterial pathogens including *Mycoplasma pneumoniae* and *Legionella pneumophila*. Most atypical pneumonia is mild and self-limited; however, for disease requiring hospitalization, consider diagnostic testing and treatment as above.

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Emergency Department Pathway for Dehydration Due to Gastroenteritis

Criteria:

Child with moderate to severe dehydration due to vomiting and diarrhea

Decreased urine output

*Mild dehydration benefits from oral rehydration better than IV

Exclude:

- Underlying metabolic disorders
- Dehydration due to surgical issues or neurologic issues (i.e.: head trauma)

Assess Severity

If child has:

- 1 feature = mild dehydration
- 2 features = moderate dehydration
- 3 features = severe dehydration

- Ill appearing
- Dry mucous membranes
- Absent tears
- Delayed capillary refill or mottling
(excluding ambient temperature or fever)

Mild

- Give Zofran ODT if ongoing vomiting (use cautiously under 6 mo)
- Oral challenge: small amount but frequent sips of clear fluids
- Accu-check and correct with oral or IV glucose

Moderate

- Give Zofran ODT
- Oral or IV rehydration
- Accu-check and correct with oral or IV glucose

Severe

Reassess after 1 hour

If tolerated oral intake

If failed oral intake

Discharge

Prescription for Zofran

- (Typically not under 6 m/o)
- (Provide only 24-48 supply)

Admission

- Place IV
- Obtain electrolytes, BUN/Cr (liver enzymes and lipase if indicated)
- Reglan IV/PO
- NS 20 ml/KG bolus over 30 mins.
- D5 NS to run at maintenance for 1 hour
- Retry oral intake: if failed then admit

Emergency Department Pathway for Status Epilepticus

Criteria:

- Seizure activity > 5 minutes
- Recurrent seizures and not returning to baseline mental status

If child **not** in status epilepticus, refer to Pediatric seizure pathway

Child with status epilepticus

- 
- Vital sign monitoring
 - Oxygen supplementation + open airway
 - IV access
 - Glucose POCT
 - Labs: CBC, metabolic panel, magnesium, phosphorus, tox screen, anti-epileptic drug level if indicated
- 

If no IV access:

- Ativan can be given IO
- Rectal Diazepam: (max dose 20mg)
 - 2-5 yrs: 0.5mg/kg
 - 6-11 yrs: 0.3mg/kg
 - 12+: 0.2mg/kg
- Intranasal Midazolam if >6y/o 0.3 mg/kg (max dose 10mg)

- Ativan 0.1 mg/kg IV (max 4mg)
- Order long acting antiepileptic medications now if not readily available in the ED

- 
- Re-assess: if still seizing in 5 minutes then repeat dose -Ativan
- 

- Re-assess: If still seizing in 5 minutes, give second agent anti-epileptic: (if neonate, give Pyridoxine Vitamin B6)

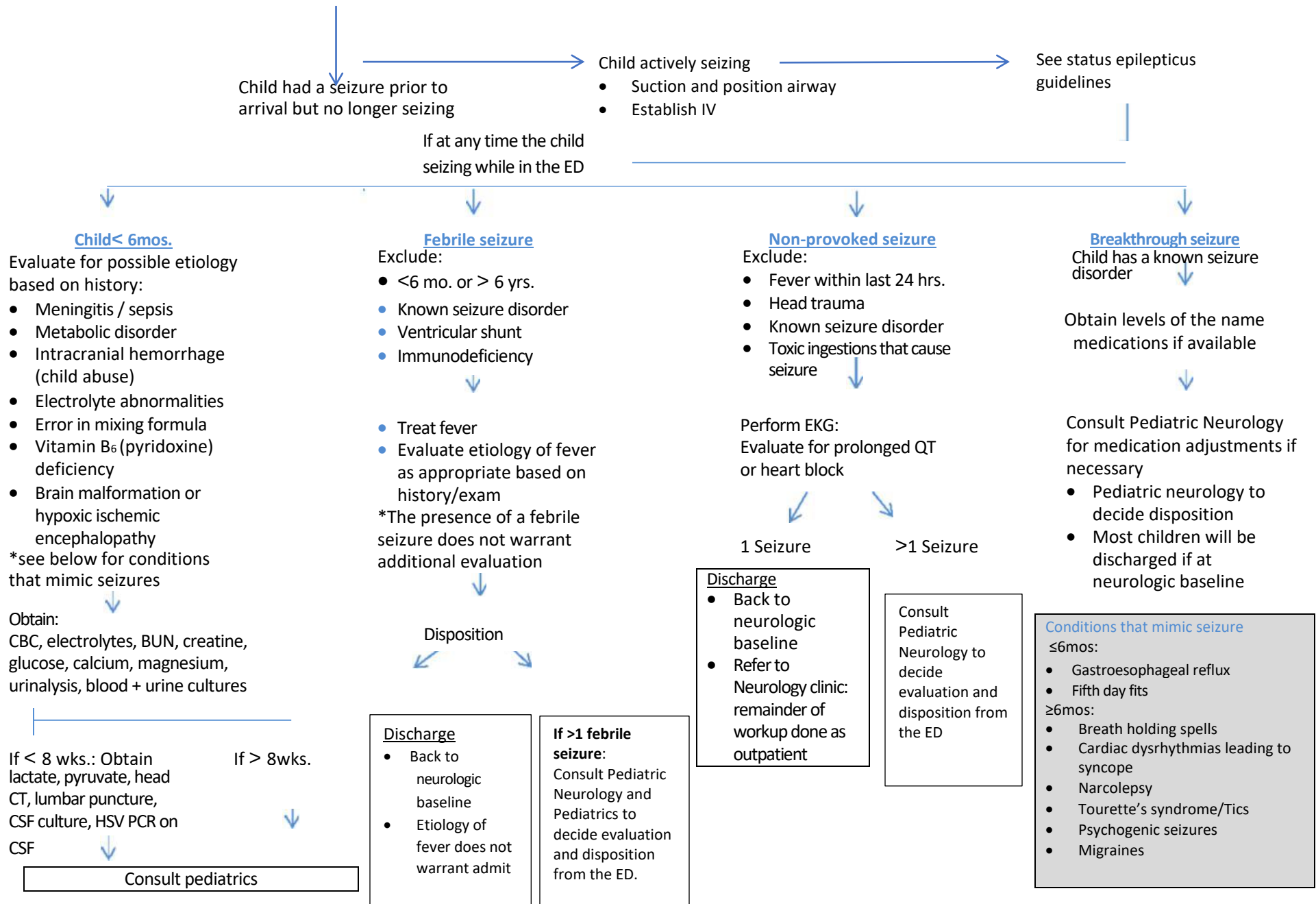
- Fosphenytoin 20 mg/kg
or
- Keppra 30-60 mg/kg
or
- Valproate 40 mg/kg

- 
- Re-assess: If still seizing in 5 minutes after infusion then give a third agent from the above list
- 

- Re-assess: If still seizing in 5 minutes after infusion then give a fourth agent from the above list

- 
- Call PICU
 - Consult with Neurology
 - Continuous EEG monitoring as needed
 - Intubate as needed
 - Additional evaluations needed:
 - CT or MRI
 - Accucheck every 1 hr

Emergency Department Pathway for Pediatric Seizure



Emergency Department Pathway – Croup

